

Aaron Ma

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EDUCATION

University of British Columbia

Bachelor of Computer Science (BCS) – Ongoing

Vancouver, BC

Sep 2023 – May 2027

University of British Columbia

Bachelor of Arts (BA) in Psychology, Minor in Commerce with Distinction

Vancouver, BC

May 2022

TECHNICAL SKILLS

Languages: HTML, CSS, JavaScript, TypeScript, Python, Java, C, C++, R, SQL

Frameworks: React, Angular, Express, Django, Flask, pytest, JUnit, Mocha, Chai, Jasmine, Karma, Jest, Playwright

Tools/Platforms: Node, Oracle, MongoDB, DynamoDB, Docker, Postman, PostgreSQL, Tableau, Swagger, AWS, Jira

Libraries: NgRx, RxJS, pandas, numPy, scikit-learn, Three.js, matplotlib, plotly, altair, recharts, Tailwindcss, D3, axios

WORK EXPERIENCE

Apera AI

Vancouver, BC

Full-Stack Software Development Engineer Intern

May 2026 – Present

- Built **React**, **TypeScript**, and **Three.js** tooling for VuePod's Automate 2026 launch, adding frame-by-frame robot playback and analytics for states, cycle times, part outcomes, and path usage
- Created and refactored **React** components and hooks, centralizing API integration, state management, validation, and error handling across simulation workflows
- Designed **Flask** REST APIs across 12 resource blueprints, centralizing authorization and error handling in reusable middleware; modeled **DynamoDB** entities and atomic provisioning for seven-resource configurations
- Scripted **DynamoDB** & **AWS S3** resource migration in **Python** between dev and production via **Cognito** OAuth, pre-signed **S3** uploads, and rollback on partial failure
- Expanded the **React** and **Flask** test suites by 80% and 35%, using **Jest**, **Playwright**, and **unittest**
- Extended a **Python** microservice that orchestrates multi-stage simulation and model training pipelines, adding a new pipeline type running end to end from user configuration through Redis workers

SS&C Technologies

Toronto, ON

Software Engineer Intern

Sep 2025 – May 2026

- Implemented **NgRx** state management, **Angular** components & services, and **RxJS** workflows for an early-stage application
- Developed and tested 20+ **Angular/TypeScript** components using **PrimeNG**, **Jasmine**, and **Karma**
- Integrated frontend components with RESTful APIs, using **Postman** and **Swagger** to validate payloads, and align API contracts
- Configured **Monaco Editor**, **Ajv**, and JSON parser utilities to support schema editing and dynamic JSON validation
- Delivered 75+ bug fixes and features across frontend and backend systems in Agile, collaborating with QA through **Jira**
- Used **AWS** CI/CD pipelines to support automated builds, testing, and deployments across application services

SAP

Vancouver, BC

Customer Success Marketing & Communications Specialist Intern

Dec 2022 – Aug 2023

- Automated data cleansing and manipulation workflows using **pandas** and **NumPy**, reducing manual processing time by 98%
- Spearheaded a Customer Success data dashboard in SAP Analytics Cloud, reducing data availability lag by 50%
- Wrote **pytest** unit tests and validation scripts to improve data quality checks with technical and non-technical documentation

PROJECTS

Netflix Wrapped – Personal

Sep 2024 – Present | [Repository](#)

- Built a React, Django, and PostgreSQL app that transforms Netflix viewing history into interactive Plotly recaps
- Processed CSV uploads asynchronously with Redis and RQ, delivering initial results while generating detailed analytics
- Created a pandas pipeline to clean, deduplicate, normalize, and aggregate viewing data by profile and year
- Developed a hybrid recommender using TF-IDF, cosine similarity, recency weighting, and user feedback to rank unseen titles
- Implemented JWT authentication with HttpOnly cookies, custom DRF authentication, CSRF protection, and Axios interceptors